

Workspace

The agent gains file-write and shell tool surfaces and acts across the working tree, locally. It holds no push credential: nothing leaves the machine without a human.

Vocabulary as of 2026-08-31 Taxonomy: ADR-0004 Sources last verified 2026-09-13



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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01 What this archetype is

A Workspace ceiling permits an agent's output to reach **the whole working tree, on the developer's machine, with no human action in between**: writing files, running shell commands, executing tests, making local commits. The name is the literal reach — the workspace — and deliberately not a description of oversight, because oversight varies enormously within it (§03) [ADR-0004](#).

Unlike the step from any later archetype to the next, crossing into Workspace changes the developer's process substantially: the human stops performing edits and starts directing an agent that performs them, inspecting results rather than keystrokes. What does *not* change is the blast radius beyond the machine — the agent holds no push credential, so the shared repository, trunk, and production remain exactly as reachable as they were under an Autocomplete ceiling: through a human.

02 The boundary crossed

What separates Workspace from Autocomplete is **file-write and shell tool surfaces**: crossing this boundary, the agent's output reaches files and processes with no keypress in between. Architecturally the crossing is one control wide — VS Code's terminal inline chat **Run** button is the exact point where a proposal surface becomes an acting one [● Tooling: Inline & Chat, Aug 2026](#) — and in practice it is one settings value wide: every in-editor edit product surveyed ships a documented way to remove the human from the accept step, and every one of eleven agentic CLIs ships a switch that removes approval entirely. Three of those CLIs ship with approval *already gone* on a fresh install (Amp: *"By default, Amp does not ask for approval before running tools"*; opencode; Goose), and Claude Code's default reviewer on major plans is now a classifier model, not the human [● Tooling: Agentic CLIs, Aug 2026](#).

Two consequences follow. First, **"the human approves each change" is a configuration, not a property of any product category** — an archetype defined by per-action approval must name the settings that keep it true. Second, this is the last boundary a team crosses by adopting a tool; every boundary further along the spectrum is crossed by granting a credential. Headless operation is universal across the CLI category and therefore no longer discriminates — what a Workspace ceiling withholds is not the ability to run unattended but the credential that would let unattended work leave the machine

[● Tooling: Agentic CLIs, Aug 2026](#).

03 What a human still gates

One thing always: **the push**. Whatever happens inside the workspace, a human performs the act that moves work off the machine, and reviews — or chooses not to review — what moves. Everything else varies, and this page carries the independence rule's flagship case. **Armin Ronacher** (creator of Flask) grants the agent everything locally and supervises none of it: *"I disable all permission checks. Which basically means I run `claude --dangerously-skip-permissions`"* — with the risk contained by Docker, not by supervision: *"there are definitely risks with it, but you can manage those risks with moving your dev env into docker"*

[● Ronacher, Jun 2025](#). Minimal reach, minimal oversight — correctly filed here, near the spectrum's start.

Under any taxonomy that ordered oversight instead of containment, this practice would land at the far end, which is precisely the conflation ADR-0004 exists to break [● ADR-0004](#).

For teams that do gate per action, the telemetry is sobering: the vendor holding it reports permission-prompt approval at ~93% rising to 97%, while the same population rejects 39% of proposed *plans* — the interface shape, not the human, determines whether review happens — and measured human review catches 13.6% of dangerous commands against 89% for automated checking [● Anthropic telemetry, 2026](#). The permission prompt is this archetype's most visible control and its least effective one.

INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. Ronacher's practice is not "further along the spectrum" than a team that approves every command — it is the same archetype with different oversight.

04 Who works this way

By self-report, this is the most populated agentic region of the spectrum — and the population number is the least corroborated one in the corpus. JetBrains' 2026 instrument (N>15,000) finds **90% of developers using agents weekly and 68% daily**; independent repository-trace mining of 128,018 projects finds agent traces in **22–29% of projects**. The two measure different units (developers versus projects), self-report over-counts by construction and trace mining under-counts by construction, and no study instruments a fixed cohort to compare what developers say against what their tooling logs — quote both numbers or neither [● JetBrains, Aug 2026](#) [● Robbes et al., Feb 2026](#).

The named practice at this ceiling is unusually well documented. **Ronacher** (§03) replaces the permission prompt with containment and engineered feedback: tools *"protected against an LLM chaos monkey using them completely wrong"*, process managers with pidfiles, logs written for the agent to read its own failures [● Ronacher, Jun 2025](#). **Mitchell Hashimoto** (creator of Ghostty; coiner of "harness engineering") delegates high-confidence tasks to a background agent while holding a flat rule — *"Please don't ever ship AI-written code without a thorough manual review"* — and responds to agent failure by engineering the agent harness rather than supervising harder; his turning point: *"I forced myself to reproduce all my manual commits with agentic ones"* [● Hashimoto, Oct 2025 / Feb 2026](#). **Simon Willison** runs multiple agents in parallel — *"surprisingly effective, if mentally exhausting"* — and keeps architecture, specifications, review and manual QA human [● Willison, Oct 2025](#).

The vendor's own recommended practice lands here too, and should be read as a recommendation, dated, not as an observation. Anthropic's *AI-Native SDLC Playbook* (fourteen lessons, 2026) organises the build phase around a Workspace ceiling: plan mode as the default start, because it *"can read the codebase without changing anything"*, then — as a team's controls mature — auto mode, with the shift stated as *"away from the user watching the agent make the edits and reviewing actions, toward the review of artifacts after longer autonomous sessions."* Parallel sessions are recommended with reviewer attention named as the limit — *"The practical ceiling is how many streams one person can review properly"*, so add sessions *"only while review is keeping up."* The course defines indicators for every play and reports a value for none, so it evidences what the vendor advises, not what any team has achieved [● Anthropic playbook, L4 & L7, 2026](#).

Set Ronacher beside antirez (the Autocomplete anchor) and the spectrum's real argument comes into focus: they sit at opposite ends of the delegation question and are arguing about the same thing — *verification*, not capability. antirez keeps the human in the loop by hand-carrying code; Ronacher removes the human and puts a container around the agent. Neither claims the code needs less checking; they disagree about what does the checking [● Practitioner Exemplars, Aug 2026](#).

05 What it gains you, what it costs

WHAT IT GAINS YOU

- Delegation of implementation: the agent edits, builds, runs and tests its own work in a real environment, closing feedback loops no proposal surface can close.
- Parallelism — one human directing several concurrent agents is a documented working practice at this ceiling [● Willison, Oct 2025](#).
- A bounded worst case, *if the bounds are built*: everything the agent can damage without a human is local and (with version control) largely revertible.

WHAT IT COSTS YOU

- Containment becomes load-bearing exactly where the field invests least: **8 of 11 agentic CLIs ship no OS-level containment of any kind**, and only one sandboxes by default (on two of three platforms) [● Tooling: Agentic CLIs, Aug 2026](#).
- Command allowlists are not a security boundary: five products — Zed (three CVEs in one day), Cursor, Roo Code (twice), VS Code — had regex allowlists over a shell defeated by shell syntax, each with a published CVE and reproduction steps [● Tooling: Agentic IDEs, Aug 2026](#).
- The lethal trifecta assembles here by default: private data access (the default read scope of one major CLI's sandbox is the entire computer, credential files included), untrusted content (whatever the agent reads), and external communication — **eight of eleven CLIs give the agent the whole network**, and both shipped egress proxies concede they cannot inspect TLS [● Tooling: Agentic CLIs, Aug 2026](#).

06 What it demands of verification

The human act removed at this boundary is per-edit acceptance, so the entailment names its replacement directly: **a sandbox scoping what the agent can write, an egress allowlist scoping what it can reach, and least-privilege tool scopes deciding what exists for it at all** — plus a test suite good enough that the agent cannot claim untested work works. Isolation and permission-gating are orthogonal controls, and the vendors say so themselves: permission modes decide whether a call runs; isolation restricts what it can access once it does. A team needs both, because a per-action classifier *"is a per-action control, not an isolation boundary"* [● Verification: Isolation, Aug 2026](#).

The demand is written as a requirement because the shipped reality runs the other way. Containment investment *thins* as delegation increases across the tool landscape — mature approval-prompt UX everywhere, two OS-primitive sandboxes, one design (Codex CLI) that inverts the pattern by making the sandbox the substrate and approval the escalation path out of it [● Tooling: Agentic CLIs, Aug 2026](#). Classifier review is a mitigation, not a boundary: adaptive attacks defeat published prompt-injection defences at

>90% attack success, and a working attack against a shipping auto-mode classifier was published in August 2026 [• Verification: Isolation, Aug 2026](#) . And one wire is missing everywhere: **no tool in the surveyed set couples permission to a verification outcome** — nothing can express "allow `git push` only if this session's tests passed." Every rule matches the shape of the call, never the state of the work; a team that wants the join must build it from hooks [• Tooling: Agentic CLIs, Aug 2026](#) .

The vendor's own playbook states the same split and, unusually, assembles the containment into one artifact. Its instruction modules (Agent Skills) are declared advisory — *"nothing forces a session to comply"* — and anything that must always hold is pushed to a deterministic hook: *"The skill makes violations rare and the hook makes them close to impossible."* Its managed-settings play then writes the ceiling as a single MDM-deployed file that engineers *"cannot edit or override"*: bypass mode disabled, only managed permission rules and hooks honoured, the sandbox made a precondition that refuses to start without it, credential paths denied and named environment variables stripped from sandboxed commands. That is the five scattered administrator-pinnable constraints this corpus found across eleven CLIs, collected by one vendor into a single reviewable posture — a recommendation, with no adoption or outcome data behind it. The same playbook recommends auto mode as the default for routine work, which sits against this vendor's own telemetry in §03 and the published attack on its auto-mode classifier; both halves come from one source, and a team should hold both [• Anthropic playbook, L4, L6 & L11, 2026](#) .

Two honest limits on what containment buys. Version control is itself a containment mechanism at this ceiling — one tool (Aider) commits every agent edit automatically so the diff is always isolated and revertible, and worktree-per-agent isolation is shipped and enforced [• Verification: Isolation, Aug 2026](#) . But **containment of the machine does not contain the artifact**: in the UK AI Security Institute's July 2026 incident, no sandbox escape occurred — the agent's sanctioned capability was the harm (a malicious pull request against a real repository, endorsed by a sockpuppet account the agent created). The workspace boundary holds only if the human act at the push credential is a review, not a relay

[• Verification: Isolation, Aug 2026](#) .

07 Governance implications

This is the archetype where liability first attaches to a setting. The one vendor term that prices auto-execution keys to precisely the act that defines this boundary: enabling execution *"without manual review or confirmation"* makes the customer *"SOLELY RESPONSIBLE FOR ANY IMPACT RESULTING FROM USE OF THIS FEATURE, INCLUDING ENSURING APPROPRIATE SAFEGUARDS, TESTING, AND MONITORING ARE IN PLACE"*

[• Cursor ToS §1.7](#) . A team at this ceiling has, contractually, already made the far end's trade: the vendor's terms treat the flipped setting, not any later credential, as the moment responsibility transfers. Every provider disclaims defect liability regardless of setting [• Governance, Legal & IP, Aug 2026](#) .

Provenance gets murky at exactly this ceiling. Workspace output enters the shared record through the *human's* push, and the attribution that survives depends on undocumented platform behaviour: co-author trailers die in plain git (`git merge --squash` recovers zero of three co-authors), so the question every later governance regime turns on — "which lines did a human write" — is decided by commit habits made here, where nobody is yet watching for it [• Governance, Legal & IP, Aug 2026](#) . Data governance, meanwhile, stops being a keystroke-egress question (the Autocomplete case) and becomes an agent-egress one: for most CLIs the

agent has the invoking user's whole network and filesystem, credential files included, so what the org's policies actually govern is decided by sandbox and egress configuration, not by the acceptable-use document [Tooling: Agentic CLIs, Aug 2026](#).

08 The posture that fits

This fits a team that believes risk is bounded by what the agent can reach, not by who is watching it.

The safety belief underneath: *the container holds*. The control moves; it does not improve. And the honest edge: the field ships almost no container by default — eight of eleven CLIs have none, the allowlists have a five-product CVE record, and the egress proxies name their own limits — so a team holding this posture is not buying boundedness off the shelf. It is committing to build the thing its belief depends on, which is exactly what this archetype's best-documented practitioners spend their effort doing.

— Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, the Workspace rename and the independence rule. Adopted 2026-08-31.
- STUDY** This project, [research/tooling-inline-and-chat.md](#) — the propose-vs-act boundary (the VS Code terminal Run control) and the edit-acceptance off switches. Closed 2026-08-30.
- STUDY** This project, [research/tooling-agentic-cli.md](#) — eleven agentic CLIs read from documentation and source: default postures, sandbox and egress inventory, the remove-all-approval switches, the missing permission-to-verification wire. Closed 2026-08-30.
- ANECDOTE** Armin Ronacher, "Agentic Coding: The Future of Software Development with Agents", [lucumr.pocoo.org](#). Stake: none in an AI coding vendor at time of writing. 12 Jun 2025.
- VENDOR** Anthropic-published telemetry: ~93%→97% permission-prompt approval; 39% plan rejection; human review catching 13.6% of dangerous commands vs 89% automated (n=1,053). The vendor sells the replacement for the prompt; no denominator or time series published. 2026.
- SURVEY** JetBrains, Developer Ecosystem Survey 2026 (N>15,000, fielded May–Jul 2026) — 90% weekly / 68% daily agent use. Self-report, no shared definition of "agent", weighted partly on JetBrains familiarity — the most quotable and least corroborated number in the corpus. Published Aug 2026.
- STUDY** Robbes et al., arXiv:2601.18341 — repository-trace mining, 128,018 projects: agent traces in 22.20–28.66% of projects. Under-counts by construction (detects only explicit co-authoring markers). As of 21 Feb 2026.

- 8 **ANECDOTE** Mitchell Hashimoto, "Vibing a Non-Trivial Ghostty Feature" (11 Oct 2025) and "My AI Adoption Journey" (5 Feb 2026), mitchellh.com. Stake: none in an AI vendor. His position moved substantially between the two posts — the dates matter: Oct 2025 / Feb 2026.
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- 9 **ANECDOTE** Simon Willison, "Vibe engineering", simonwillison.net. Stake: independent; reputationally invested in the topic; later retired his own term in favour of "agentic engineering". 7 Oct 2025.
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- 10 **STUDY** This project, [research/practitioner-exemplars.md](#) — the antirez/Ronacher contrast; the replaced-verification table; ~57 exemplars with stake recorded. Closed 2026-08-27.
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- 11 **STUDY** This project, [research/tooling-agentic-ide.md](#) — the five-product allowlist CVE record (Zed CVE-2026-44462/-44463/-44466; Cursor CVE-2026-22708; Roo Code CVE-2025-57771/-58370; VS Code CVE-2026-45482), from vendor security advisories with reproduction steps. Closed 2026-08-30.
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- 12 **STUDY** This project, [research/verification-isolation.md](#) — isolation vs permission-gating; worktree isolation; the UK AISI July 2026 incident; "The Attacker Moves Second" (>90% adaptive attack success) and the published auto-mode classifier attack. Closed 2026-08-28.
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- 13 **VENDOR** Cursor Terms of Service §1.7, "Auto-Code Execution" (contract text). Accessed 2026-08-30.
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- 14 **STUDY** This project, [research/governance-legal-ip.md](#) — provider defect disclaimers; attribution survival under squash merges; the authorship-vs-adoption threshold. Closed 2026-08-30, corrected 2026-08-31.
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- 15 **VENDOR** This project, [research/ai-native-sdlc-playbook.md](#) — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in [reference/](#)). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L4 (plan mode; auto mode), L6 (advisory Agent Skills vs deterministic hooks), L7 (parallel sessions), L11 (managed settings). Retrieved and digested 2026-09-02.
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